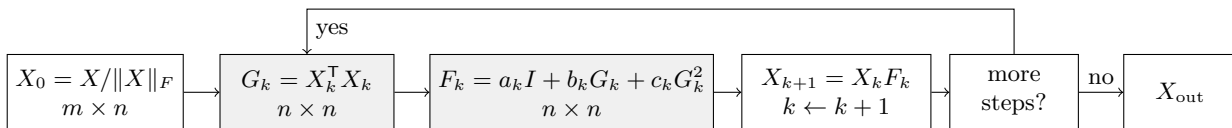
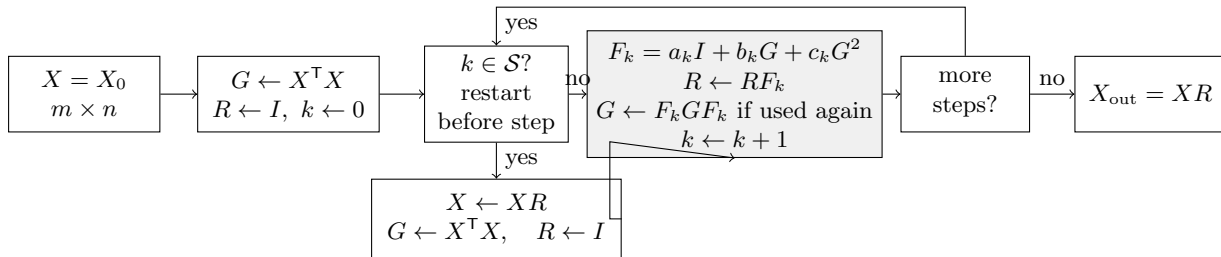


Standard Newton–Schulz: materialize the tall matrix every iteration



Gram Newton–Schulz: accumulate in Gram space and restart explicitly



Shown for $m \geq n$. For $m < n$, use $G = XX^\top$ and apply every factor on the left.